Homework 1 Conceptuals: Beras Pt. 1

CS 1470/2470: Deep Learning

Due February 9, 2024 at 6:00pm EST

Answer the following questions, showing your work where necessary. Please explain your answers and work. *Please do NOT include your name anywhere within this submission. Points will be deducted if you do so.*

1 Conceptual Questions

1. Consider Mean-Square Error (MSE) loss. Why do we square the error? **Please provide two reasons**. (*2-3 sentences*)

   Solutions: (1)Errors can be positive or negative, depending on whether the predicted value is higher or lower than the actual value. If we were to simply sum these errors without squaring them, positive and negative errors could cancel each other out, giving a misleading indication that the model is performing well when it might not be. Squaring ensures that all errors are positive, providing a true measure of the model's performance.

   (2)The MSE loss function is continuously differentiable, making it a smooth function for optimization. This is crucial for gradient descent algorithms, which rely on calculating the gradient of the loss function to update model parameters. The squared term ensures that the gradient is continuous and defined everywhere, allowing for more reliable and stable optimization.

2. We covered a lemonade stand example in the class (Intro to Machine Learning), which was a regression task.

   (a) How would you convert it into a classification task?

   (b) What are the benefits of keeping it a regression task? What are the benefits of converting it to a classification task?

   (c) Give another example of a classification and a regression task each.

   Solutions: a. To convert the task of using the day's temperature to predict the profit of selling lemonade from a regression to a classification task, we would categorize the profit into discrete classes rather than predicting the exact profit value. For instance, we could classify days as "high profit," "medium profit," and "low profit" based on temperature ranges or profit thresholds.

b.Benefits of keeping it a regression task would be that regression allows for the prediction of precise profit amounts, offering more detailed insights into expected earnings from lemonade sales on a given day. This precision can aid in more accurate financial planning and inventory management.

Benefits of converting it to a classification task would be that classification simplifies the prediction problem by focusing on broader categories of outcomes rather than exact values. This can make the model more robust to noise and less sensitive to small fluctuations in temperature that may not have a significant impact on profit categories.

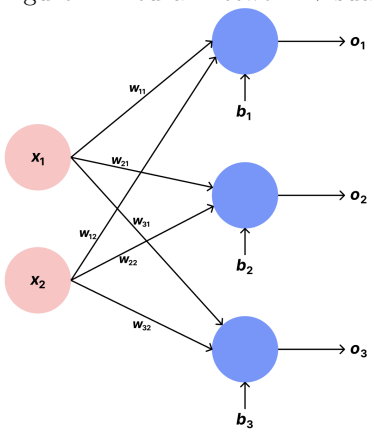
c. Another example can be models as regression or classification task are customer churn,which refers to customers discontinuing their use of a service:

Modeled as a regression task: The objective is to estimate the probability of a customer churning within a given time frame, such as the next month.The output is a continuous value between 0 and 1 representing the probability of churn. The benefits of modeling this as a regression task would be providing a nuanced view of churn risk, allowing businesses to prioritize interventions based on the estimated risk level.

Modeled as a classification task:The objective would be classifing customers into two categories, such as "Will Churn" or "Will Not Churn," based on their characteristics and interaction with the service. The output is a categorical label indicating the predicted class. The benefits of moedeling this as a classification task would be simplifing decision-making processes by clearly segmenting customers into actionable groups. This binary classification can be directly used to implement retention strategies or further analysis.

3. Consider the following network:

Figure 1: Neural Network Visual



Let the input, weight, bias, and expected hypothesis matrix $o_{expected}$ be the following:

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad w = \begin{bmatrix} 0.4 & 0.7 & 0.1 \\ 0.2 & 0.4 & 0.5 \end{bmatrix} \quad b = \begin{bmatrix} 0.2 \\ 0 \\ 0.4 \end{bmatrix} \quad o_{expected} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Answer the following questions, write out equations as necessary, and **show your work for every step**.

(a) Complete a manual forward pass (i.e. find the output matrix o AKA $\hat{y}$). Assume the network uses a linear function for which the above matrices are compatible.

(b) Using Mean Squared Error as your loss metric, compute the loss relative to $o_{expected}$.

(c) Perform a single manual backward pass assuming that the the expected output $o_{expected}$, AKA the ground truth value y, is given. Calculate the partial derivative of the loss with respect to each parameter.

(d) Give the updated weight and bias matrices using the result of your gradient calculations. Use the stochastic gradient descent algorithm with learning rate = 0.1.

Solutions

a) Manual Forward Pass

Given the input matrix x, the transposed weight matrix w^T, and the bias vector b, the output matrix o (or $\hat{y}$) is computed as follows:

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad w^T = \begin{bmatrix} 0.4 & 0.2 \\ 0.7 & 0.4 \\ 0.1 & 0.5 \end{bmatrix}, \quad b = \begin{bmatrix} 0.2 \\ 0 \\ 0.4 \end{bmatrix}$$

$$o = w^T \cdot x + b = \begin{bmatrix} 1.0 \\ 1.5 \\ 1.5 \end{bmatrix}$$

b) Compute the Loss

The MSE loss relative to the expected output $o_{expected}$ is:

$$o_{expected} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (o_{expected_i} - o_i)^2 = 1.25$$

c) Manual Backward Pass

The partial derivatives of the loss with respect to the output o are calculated as follows:

$$\frac{\partial \mathrm{L}}{\partial o} = 2 \cdot \frac{1}{n}(o - o_{expected}) = \begin{bmatrix} 0 \\ -0.5 \\ 1.5 \end{bmatrix}$$

The output o is calculated as $o = w^T \cdot x + b$. Therefore, the derivative of o with respect to w involves the input x, since for a linear function, this derivative is simply the input x itself.

Combining the above with the chain rule, we get:

$$\frac{\partial \mathrm{L}}{\partial w} = \frac{\partial \mathrm{L}}{\partial o} \cdot \frac{\partial o}{\partial w} = 2 \cdot \frac{1}{n} \cdot x^T \cdot (o - o_{expected}) = \begin{bmatrix} 0 & 0 \\ -0.5 & -1 \\ 1.5 & 3 \end{bmatrix}$$

Since the bias b is added directly to the weighted sum, its derivative in the output equation is 1. Thus:

$$\frac{\partial \mathrm{L}}{\partial b} = \frac{\partial \mathrm{L}}{\partial o} \cdot 1 = 2 \cdot \frac{1}{n} \cdot (o - o_{expected}) = \begin{bmatrix} 0 \\ -0.5 \\ 1.5 \end{bmatrix}$$

d) Parameter Update using Stochastic Gradient Descent

Given a learning rate $\alpha = 0.1$, the update rule for each parameter is:

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial \mathrm{L}}{\partial w} = \begin{bmatrix} 0.4 & 0.2 \\ 0.7 & 0.4 \\ 0.1 & 0.5 \end{bmatrix} - 0.1 * \begin{bmatrix} 0 & 0 \\ -0.5 & -1 \\ 1.5 & 3 \end{bmatrix} = \begin{bmatrix} 0.4 & 0.2 \\ 0.75 & 0.3 \\ -0.05 & 0.2 \end{bmatrix}$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{\partial \mathrm{L}}{\partial b} = \begin{bmatrix} 0.2 \\ 0 \\ 0.4 \end{bmatrix} - 0.1 * \begin{bmatrix} 0 \\ -0.5 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.05 \\ 0.25 \end{bmatrix}$$

4. Empirical risk $\hat{R}$ for another machine learning model called logistic regression (used widely for classification) is:

$$\hat{R}_{\log}(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} \ln\big(1 + \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big)\big)$$

This loss is also known as log-loss. Take the gradient of this empirical risk w.r.t $\mathbf{w} \in \mathbb{R}^d$. Your answer may need to include parameters $\mathbf{w}$, number of examples n, labels $y_i \in \mathbb{R}$, and training examples $\mathbf{x}_i \in \mathbb{R}^d$. Show your work for every step!

Solutions:

Given the empirical risk for logistic regression as:

$$\hat{R}_{\log}(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} \ln\big(1 + \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big)\big),$$

we aim to compute the gradient with respect to the weight vector $\mathbf{w}$.

The gradient of a single term in the sum with respect to $\mathbf{w}$ is computed as follows:

$$\frac{\partial}{\partial \mathbf{w}} \ln\big(1 + \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big)\big).$$

Using the chain rule, this becomes:

$$\frac{1}{1 + \exp(-y_i \mathbf{w}^\top \mathbf{x}_i)} \cdot \frac{\partial}{\partial \mathbf{w}} \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big).$$

The derivative of $\exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big)$ with respect to $\mathbf{w}$ is:

$$-y_i \mathbf{x}_i \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big),$$

leading to the expression:

$$\frac{-y_i \mathbf{x}_i \exp\big(-y_i \mathbf{w}^\top \mathbf{x}_i\big)}{1 + \exp(-y_i \mathbf{w}^\top \mathbf{x}_i)}.$$

This simplifies to:

$$-y_i \mathbf{x}_i (1 - \sigma(y_i \mathbf{w}^\top \mathbf{x}_i)),$$

where $\sigma(z) = \frac{1}{1+\exp(-z)}$ is the logistic function.

Averaging these gradients over all n samples gives the gradient of the empirical risk:

$$\nabla_{\mathbf{w}} \hat{R}_{\log}(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} -y_i \mathbf{x}_i (1 - \sigma(y_i \mathbf{w}^\top \mathbf{x}_i)).$$

2 2470-only Questions

The following are questions that are only required by students enrolled in CSCI2470. **For each question, include a 2-4 sentence argument for why this mathematical detail may be considered interesting and useful.**

1. You have two interdependent random variables $x, y \sim X, Y$ and you'd like to learn a mapping $f : X \to Y$. The best function you can get is $f(x) = \mathbb{E}[Y|X = x]$. This does not account for noise associated with random events. Let ξ be the noise such that $y = f(x) + \xi$. Prove that $\mathbb{E}[\xi] = 0$.

   Solutions: Let $y = f(x) + \xi$, where y is the realization of the random variable Y, $f(x)$ is the expected value of Y given $X = x$, and ξ represents the noise. Our goal is to prove that $\mathbb{E}[\xi] = 0$.

   First, we express ξ in terms of y and $f(x)$:

   $$\xi = y - f(x)$$

   Applying the linearity of expectation, we get:

   $$\mathbb{E}[\xi] = \mathbb{E}[y - f(x)]$$

   Since expectation is linear, we can further break it down into:

   $$\mathbb{E}[\xi] = \mathbb{E}[y] - \mathbb{E}[f(x)]$$

   Next, the law of total expectation states that if X and Y are two random variables, then:

   $$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$$

   Given $f(x) = \mathbb{E}[Y|X = x]$, it follows that:

   $$\mathbb{E}[f(x)] = \mathbb{E}[\mathbb{E}[Y|X]]$$

   Therefore, by applying the law of total expectation, we see that $\mathbb{E}[f(x)] = \mathbb{E}[Y]$. Substituting this back into our expression for $\mathbb{E}[\xi]$, we find:

   $$\mathbb{E}[\xi] = \mathbb{E}[Y] - \mathbb{E}[Y] = 0$$

   Why this mathematical detail may be considered interesting and useful? This proof demonstrates that the expectation of the noise term ξ in a linear model where $y = f(x) + \xi$ is zero. This indicates that while individual observations of Y may deviate from $f(x)$ due to ξ, these deviations do not have a systematic bias and average out to zero across the distribution of X.

2. Let $p_{\mathbf{w}}$ conditionally parameterize a normal distribution such that:

$$p_{\mathbf{w}}(y|\mathbf{x}) \sim \mathcal{N}(\mathbf{x}^T\mathbf{w}, 1)$$

Show that the maximizing likelihood objective can be written by using $\arg\min_{\mathbf{w}}$ and the squared error between $\mathbf{x}^T\mathbf{w}$ and y.

Intuitively, this is a probabilistic interpretation of linear regression. We are interested in the joint probability of how the behaviour of the output y is conditional on the values of the input vector $\mathbf{x}$, as well as any parameters of the model $\mathbf{w}$. The model is represented as a Gaussian distribution (prior) with the linear function output as the mean. We assume that the variance in the model is fixed ($= 1$) (i.e. that it doesn't depend on $\mathbf{x}$).

Solutions

The likelihood of observing y given $\mathbf{x}$ and model parameters $\mathbf{w}$ under our normal distribution assumption is:

$$L(\mathbf{w}|\mathbf{x}, y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y - \mathbf{x}^T\mathbf{w})^2\right)$$

Taking the logarithm of the likelihood, we obtain the log-likelihood:

$$\log L(\mathbf{w}|\mathbf{x}, y) = \log\left(\frac{1}{\sqrt{2\pi}}\right) - \frac{1}{2}(y - \mathbf{x}^T\mathbf{w})^2$$

Since $\log\left(\frac{1}{\sqrt{2\pi}}\right)$ is a constant, maximizing the log-likelihood with respect to $\mathbf{w}$ is equivalent to minimizing the squared error term:

$$-\frac{1}{2}(y - \mathbf{x}^T\mathbf{w})^2$$

Therefore, the maximizing likelihood objective can be written as minimizing the squared error between the model prediction and the observed output:

$$\arg\min_{\mathbf{w}} \frac{1}{2}(y - \mathbf{x}^T\mathbf{w})^2$$

Why this mathematical detail may be considered interesting and useful? This derivation shows that in the context of linear regression modeled as a Gaussian distribution with fixed variance, maximizing the likelihood of the model parameters is equivalent to minimizing the squared error between the predicted outputs and the observed targets. This provides a probabilistic interpretation of linear regression, linking it to the principle of maximum likelihood estimation.

3. Assume instead that each element $w_i \sim \mathcal{N}(0, \frac{1}{\lambda})$ and $p_{\mathbf{w}}(y\ |(\mathbf{x}, \mathbf{w})) \sim \mathcal{N}(\mathbf{w}^T\mathbf{x}, 1)$. Show that the following must be true:

$$\arg\max_{\mathbf{w}} P(\mathbf{w}) \prod_{i=1}^{n} p_w(y_i|\mathbf{x}_i) = \arg\min_{\mathbf{w}} \left\{ \frac{1}{2} \sum_{i=1}^{n} (\mathbf{w}^T\mathbf{x}_i - y_i)^2 + \frac{\lambda}{2} \sum_{j=1}^{d} w_j^2 \right\}$$

This is a *regularized version* of linear regression (called ridge regression where the second term tries to push the weight values that are least influential to 0. λ is a hyperparameter (selected by user).

Solutions

The prior distribution for each weight w_i is given by:

$$P(w_i) \sim \mathcal{N}(0, \frac{1}{\lambda})$$

The likelihood of observing y_i given $\mathbf{x}_i$ and $\mathbf{w}$ is:

$$p_{\mathbf{w}}(y_i|\mathbf{x}_i) \sim \mathcal{N}(\mathbf{w}^T\mathbf{x}_i, 1)$$

The posterior probability of $\mathbf{w}$ given the data is proportional to the product of the prior and the likelihood:

$$P(\mathbf{w}|\{\mathbf{x}_i, y_i\}_{i=1}^n) \propto P(\mathbf{w}) \prod_{i=1}^{n} p_w(y_i|\mathbf{x}_i)$$

Taking the logarithm of the posterior and dropping terms that do not depend on $\mathbf{w}$, we aim to maximize:

$$\log P(\mathbf{w}) + \sum_{i=1}^{n} \log p_w(y_i|\mathbf{x}_i)$$

Substituting the expressions for the prior and likelihood, we get:

$$-\frac{\lambda}{2} \sum_{j=1}^{d} w_j^2 - \frac{1}{2} \sum_{i=1}^{n} (\mathbf{w}^T\mathbf{x}_i - y_i)^2$$

Maximizing the above expression with respect to $\mathbf{w}$ is equivalent to minimizing its negation:

$$\arg\min_{\mathbf{w}} \left\{ \frac{1}{2} \sum_{i=1}^{n} (\mathbf{w}^T\mathbf{x}_i - y_i)^2 + \frac{\lambda}{2} \sum_{j=1}^{d} w_j^2 \right\}$$

Why this mathematical detail may be considered interesting and useful? This derivation shows that the ridge regression objective, which includes a regularization term to penalize large weights, can be obtained from a probabilistic model with a Gaussian prior on the weights and a Gaussian likelihood for the outcomes. The regularization term controlled by λ encourages the model to prefer solutions with smaller weight magnitudes, thereby mitigating overfitting.